

ADVANCED STRENGTH OF MATERIALS

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.**This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. Answer the following questions (any ten):

1×10=10

Fill in the blanks:

- i) Bending stress at N.A. is _____.
- ii) The section modulus of a beam having width b and depth d is given by _____.
- iii) The ratio of the polar moment of inertia to the radius of the shaft is called _____.
- iv) Force required to deflect a spring through unit deflection is called _____.
- v) In a strained body, maximum number of principal planes may be _____.
- vi) The maximum amount of strain energy that can be stored in a body without deformation is called _____.
- vii) A shaft is said to be in pure in pure tension, if it is subjected to _____ only.

State whether the following statements are True or False:

- viii) The stiffness of spring is also called spring constant.
- ix) The product of $G \times I_p$ is called torsional rigidity.
- x) The neutral axis of a beam is subjected to maximum tensile stress.
- xi) Core or kernel is the area within which the resultant of the load may be applied to avoid tensile stress.
- xii) Polar moment of inertia of a circular section of diameter D is $\pi/32 D^4$
- xiii) If section modulus of a beam is increase, the bending stress will increase.
- xiv) For a beam with rectangular section, maximum shear stress occurs at neutral axis.
- xv) If a coil spring is cut into two halves, deflection of the spring will be half.

2. What is simple bending? What are the assumptions made in the simple bending? 2+3
3. A steel plate 12 mm thick is bent around a drum. If the maximum stress induced in the plate due to bending be 400 N/mm^2 , find the radius of the drum. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 5
4. Strain energy of two elastic bars subjected to gradual loading having cross-sectionals area are 400 mm^2 and 800 mm^2 are same. Establish a relation between stresses of two bars. 5
5. An I-section beam of flanges $20 \text{ cm} \times 2 \text{ cm}$ and web $30 \text{ cm} \times 1 \text{ cm}$ is acted on by a shear force of 150 kN determine: i).the maximum and minimum shearing stress in the web. ii).the maximum shear stress in the flange. 5
6. The principal stress at a point in the section of a boiler shell are 80 N/mm^2 and 40 N/mm^2 . Find the normal, tangential and resultant stress across a plane through the point inclined at 50° to the plane carrying 80 N/mm^2 stress. 5

7. Deduce torsion equation i.e. $\frac{T}{I_p} = \frac{\sigma_s}{R} = \frac{G\theta}{L}$ for solid shaft. 5
8. A solid shaft of 80 mm diameter is to be replaced by a hollow shaft of external diameter of 100 mm. 5 Determine the internal diameter of hollow shaft if the same power is to be transmitted by the both shafts at the same angular velocity and shear stress. 5
9. A closed coiled helical spring having the mean coil diameter is 12 times than the wire diameter. The spring is to absorb 30000 N cm of energy with an extension of 15 cm. The maximum shear stress set up in the spring is not to exceed 10000 N/cm². Determine – (a) mean coil diameter, (b) number of turns. 5
-

<https://www.wbsctonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

March 2021

ADVANCED STRENGTH OF MATERIALS

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any ten): 1x10

State whether the following statements are True or False:

- i) The strength of the beam mainly depends on the bending moment.
- ii) Strain energy is set up in elastic material.
- iii) Moment of resistance of beam is independent of its section modulus.
- iv) If the section modulus of a beam is increased, the bending stress of the beam will be decrease.
- v) Proof resilience is the maximum strain energy upto elastic limit.
- vi) Spring index is the ratio of mean coil diameter to the diameter of spring.
- vii) For a circular section shear stress distribution is parabola.
- viii) The unit of section modulus and area are same.

Fill in the blanks:

- ix) The bending equation is _____.
- x) The bending stress on a beam section is maximum at _____.
- xi) For a T-section beam, maximum shear stress occurs at _____.
- xii) The angle between the principle planes is _____.
- xiii) The radius of Mohr's stress circle indicates maximum _____ stress.
- xiv) Strain energy is set up in _____ material.
- xv) Proof resilience per unit volume is called _____.
- xvi) The unit of modulus of elasticity is same as of _____.

2. An I-section beam of flanges 20 cm × 2 cm and web 30 cm × 1 cm is acted on by a shear force of 150 kN determine: i).the maximum and minimum shearing stress in the web. ii).the maximum shear stress in the flange. 5
3. The principal stress at a point in the section of a boiler shell are 80 N/mm² and 40 N/mm². Find the normal, tangential and resultant stress across a plane through the point inclined at 50° to the plane carrying 80 N/mm² stress. 5
4. Deduce torsion equation i.e. $\frac{T}{I_p} = \frac{\sigma_s}{R} = \frac{G\theta}{L}$ for solid shaft. 5
5. A solid shaft of 80 mm diameter is to be replaced by a hollow shaft of external diameter of 100 mm. Determine the internal diameter of hollow shaft if the same power is to be transmitted by the both shafts at the same angular velocity and shear stress. 5
6. A closed coiled helical spring having the mean coil diameter is 12 times than the wire diameter. The spring is to absorb 30000 N cm of energy with an extension of 15 cm. The maximum shear stress set up in the spring is not to exceed 10000 N/cm². Determine – (a) mean coil diameter, (b) number of turns. 5
7. What is simple bending? What are the assumptions made in the simple bending? 2+3

8. A steel plate 12 mm thick is bent around a drum. If the maximum stress induced in the plate due to bending be 400 N/mm^2 , find the radius of the drum. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 5
9. Strain energy of two elastic bars subjected to gradual loading having cross-sectional area are 400 mm^2 and 800 mm^2 are same. Establish a relation between stresses of two bars. 5
-

<https://www.wbscteonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

STRENGTH OF MATERIALS*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten): 1 x 10
- i. Spring is used to- a) Absorb shock, b) Accelerate speed of a vehicle, c) Both (a) & (b), d) None of the above.
 - ii. The property of a material by virtue of which it can be rolled into plate known as-
a) Ductility b) toughness c) malleability d) none of these.
 - iii. Neutral axis passes through-
a) The C.G. of the section of the beam b) the top layer of the beam
c) The bottom layer of the beam d) none of the above
 - iv. For circular section of diameter 'd', the section modulus is given by _____.
 - v. If diameter of a long column is reduced by 20%, the percentage reduction of Euler buckling load is -
(a) 59, (b) 49, (c) 8, (d) 36.
 - vi. Slope and deflection of a beam can be determined by-
a) Double integration method b) Moment area method
c) Macaulay's method d) any of the above
 - vii. A strut is a member which carries - (a) tensile load, (b) shear load, (c) compressive load, (d) both tensile and compressive load.
 - viii. Poisson's ratio for any material cannot be more than-
a) 0.3, b) 0.5, c) 0.8, d) 1
 - ix. Deformation of a body under tensile load P of cross sectional area A of Length L having Young's modulus E is given by-
a) $\frac{PE}{AL}$ b) $\frac{AE}{PL}$
c) $\frac{PL}{AE}$ d) none of these
 - x. Hooke's law holds good upto the - (a) elastic limit, (b) plastic limit, (c) yield point, (d) proportional limit.
 - xi. Unit of modulus of rigidity - a) N/m^2 b) N/mm^2 c) both a and b d) none of these
 - xii. Bending moment diagram of uniformly distributed force will be-
a) Linear b) Inclined
c) Non-linear d) Horizontal
 - xiii. Percentage of elongation is the measure of-
a) Ductility, b) Malleability, c) Plasticity, d) Elasticity
 - xiv. If diameter of a shaft is doubled the power transmitted capacity will be - (a) same, (b) either twice or half, (c) four times, (d) eight times.

- xv. Shafts are designed on the basis of-
(a) strength (b) rigidity (c) either of the above (d) either (a) or (b)

2. Fill in the blanks (any ten):

1x10

- i. The maximum bending moment (M) caused by the total uniformity distributed load (W) over a simply supported span (L) is given by _____.
- ii. The ratio of crippling load to working load is called _____.
- iii. The angle of twist is _____ proportional to the twisting moment
- iv. Reaction at free end of a cantilever beam is _____.
- v. A beam can be made of uniform strength by keeping the _____ uniform, and varying the _____.
- vi. Unit of radius of gyration is _____.
- vii. The ratio of lateral strain to linear strain is known as _____.
- viii. When a SF at a point is zero, then BM at that point is _____.
- ix. Permanent set occurs when stress is beyond _____ limit.
- x. For a beam with rectangular section, the ratio of maximum shear stress to average shear stress is _____.
- xi. Unit of coefficient of thermal expansion is _____.
- xii. _____ is a graphical method to determine various stresses on an oblique plane of a strained material.
- xiii. The combined effect of external forces acting on a body is called a _____.
- xiv. The strain at right angles to the direction of applied load is known as _____.
- xv. Percentage elongation is a measure of _____.

3. Answer the following questions (any ten):

1x10

- i. Define brittle material.
- ii. What is the helix angle of helical spring?
- iii. Show uniformly varying load on a beam.
- iv. Write the mathematical expression of Temperature stress.
- v. Write down the relation between bulk modulus & compressibility.
- vi. What do you mean by polar modulus?
- vii. What is the expression of Torque transmitted by a solid shaft of diameter d?
- viii. Write the mathematical expression of Polar Modulus of a shaft.
- ix. What is torsional rigidity?
- x. What is the maximum value of S.F, in case of simply supported beam of span l loaded with a point load W at its mid span?
- xi. Describe bending stress.
- xii. What is meant by section modulus?
- xiii. What is law of transmissibility?
- xiv. What is the factor of safety?
- xv. What is SI unit of UDL?

Group B

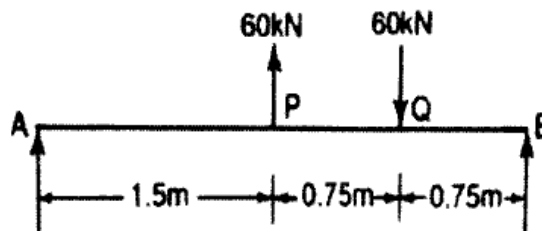
4. Answer the following questions (any six):

2x6

- i. What is value of bulk modulus of a material having $E=250\text{GPa}$ and $G=80\text{GPa}$?
- ii. Draw a stress strain diagram for ductile materials and indicate all salient features on it.
- iii. Write the equation of torsion and give the names of each terms.
- iv. Define stiffness of a spring.
- v. Write down the difference between deflection & bending moment.
- vi. What is Pure Bending?
- vii. Define SFD and BMD.
- viii. State the perpendicular axis theorem.
- ix. What do you mean by slenderness ratio?
- x. What are the application areas of spring?

Group C

5. Answer the following question (any one): 1x6
- i. What is moment of inertia. What is the difference between mass moment of inertia & area moment of inertia? 2+4
 - ii. Discuss the stress strain behavior of ductile material. Establish the relation $E = 2G(1 + \mu)$ where E is Young's modulus, G is modulus of rigidity and μ is Poisson's ratio. 2+4
 - iii. Find the MI of a Triangular area with respect to the axis passing through the base of the triangle, where, base=b and height =h. 6
6. Answer the following question (any one): 1x6
- i. What are the assumptions made in the Euler's formula? Explain the term effective length of a column. 4+2
 - ii. A cantilever 2m long carries point loads of 4kN, 5kN, 3kN at distances 1m, 1.5m, and 2m respectively from the fixed end sides. Draw the shear force and bending moment diagram for the beam. 6
 - iii. A T beam 200mm*150mm*10mm is used over a span of 3 meter It carries a uniformly distributed load of 12000 N per meter run. Find the position of the neutral axis and the maximum stress induced in the beam due to bending. Also state the nature of maximum bending stress. 6
7. Answer the following question (any one): 1x6
- i. A solid circular shaft of 100 mm diameter is transmitting 120 kW at 150 r.p.m. Find the intensity of shear stress in the shaft. <https://www.wbscteonline.com> 6
 - ii. What do you mean by equivalent stiffness of helical spring? Draw sketches showing series connection and parallel connection of helical springs. 2+4
 - iii. A beam of 3m span is loaded as shown in the fig 6. Calculate the deflection at P and Q and slope at A. Take the value of $EI = 840 \text{ KN-m}^2$ 6



March 2023

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20
- i) The material which shows little or no plastic deformation is called – (a) ductile material (b) brittle material (c) malleable material (d) plastic material.
 - ii) The point at which there is increase in strain with no appreciable increase in stress is known as – (a) Breaking point (b) Yield point (c) Elastic limit (d) Proportional limit.
 - iii) Ratio of Hoop stress and longitudinal stress of a material is equal to – (a) 1 (b) $\frac{1}{2}$ (c) 2 (d) none of these.
 - iv) Temperature stress developed in a bar depends upon – (a) co-efficient of linear expansion (b) change of temperature (c) Young's modulus (d) all of these.
 - v) The area moment of inertia is often called _____ moment of area. – (a) first (b) second (c) third (d) fourth
 - vi) Moment of Inertia of a triangular lamina of height 'h' about its base 'b' is – (a) $\frac{b \times h^3}{12}$ (b) $\frac{b \times h^3}{36}$ (c) $\frac{h \times b^3}{12}$ (d) none of these.
 - vii) If I_{OX} and I_{OY} be moments of inertia of a lamina about mutually perpendicular axes OX and OY in the plane of the lamina and I_{OZ} be the moment of inertia of the lamina about an axis OZ normal to the lamina and passing through the point of intersection of the axis OX and OY then $I_{OZ} = I_{OX} + I_{OY}$. The above theorem is known as – (a) theorem of parallel axes (b) theorem of perpendicular axes (c) three moments theorem (d) none of these.
 - viii) A simple supported beam of length 10 m carries uniformly distributed load of 10 kN/m over the entire span. The magnitude of each support reaction will be – (a) 25 kN (b) 30 kN (c) 50 kN (d) 100 kN.
 - ix) The maximum bending moment of a cantilever beam appears at – (a) fixed end (b) free end (c) mid-span (d) none of these.
 - x) If a beam is loaded with UDL, the nature of bending moment diagram will be – (a) straight horizontal line (b) straight inclined line (c) parabolic (d) cubic.
 - xi) At the point of contra flexure, the value of bending moment will be – (a) maximum (b) minimum (c) zero (d) none of these.
 - xii) The bending stress at neutral axis is – (a) maximum (b) minimum (c) zero (d) tensile in nature.
 - xiii) The section modulus of a circular section of diameter d is – (a) $\frac{\pi d^2}{32}$ (b) $\frac{\pi d^3}{32}$ (c) $\frac{\pi d^3}{64}$ (d) $\frac{\pi d^4}{64}$.
 - xiv) Neutral axis passes through – (a) the top layer of the beam (b) the bottom layer of the beam (c) the CG of the beam (d) none of these.

- xv) Maximum deflection of a beam occurs where the slope of the beam is – (a) maximum (b) 45° (c) 90° (d) 0° .
- xvi) The maximum deflection of a cantilever beam appears at – (a) fixed end (b) free end (c) mid-span (d) none of these.
- xvii) Doubling the span of beam will result in increase in deflection of the beam by – (a) two times (b) four times (c) eight times (d) none of these.
- xviii) Euler's formula for crippling load is best suitable for slenderness ratio more than – (a) 50 (b) 70 (c) 80 (d) 100
- xix) The crippling load of a column depends on – (a) cross-sectional area (b) modulus of rigidity (c) slenderness ratio (d) all of these.
- xx) The ratio of actual length to the equivalent length of a column having one end fixed and other end free is – (a) 1:1 (b) $\sqrt{2}$:1 (c) 1:2 (d) $1:\sqrt{2}$.
- xxi) The strength of a hollow shaft for the same length, material and weight is _____ a solid shaft. – (a) less than (b) more than (c) equal to (d) none of these.
- xxii) In pure torsion, the shaft gets deformed due to – (a) bending moment only (b) axial load only (c) twisting moment only (d) all of these.
- xxiii) For a hollow circular shaft torsional shear stress at the centre is – (a) maximum (b) zero (c) minimum (d) none of these.
- xxiv) The predominant effect of an axial tensile force on a helical spring is – (a) bending (b) compression (c) twisting (d) tension.
- xxv) Two helical springs of stiffness K is attached in parallel. The stiffness of the equivalent system will be – (a) 2K (b) K/2 (c) 2K (d) 4K

Group B

2.
 - i) Draw the stress-strain diagram of Mild Steel and locate all the salient points in the diagram.
 - ii) A steel wire 2 m long and 3 mm in diameter is extended by 0.75 mm when a weight W is suspended from the wire. If the same weight is suspended from a brass wire, 2.5 m long and 2 mm in diameter, it is elongated by 4.64 mm. Determine the modulus of elasticity of brass if that of steel be $2.0 \times 10^5 \text{ N/mm}^2$. 3+5
3.
 - i) Define volumetric strain and bulk modulus of a material.
 - ii) A copper bar of diameter 25 mm and length 2 m is rigidly fixed at both ends. Find the thermal stress induced in the bar if the temperature is reduced by 40°C . Take $E = 100 \text{ GPa}$, $\alpha = 16 \times 10^{-6}/^\circ\text{C}$. 2+6
4.
 - i) State the "Theorem of parallel axis".
 - ii) Show that the moment of inertia of a circular section with respect to an axis passing through its centre of gravity is $\pi D^4/64$, where D is the diameter of the circular section. 2+6
5.
 - i) Write down the relationship between shear force and bending moment and explain its significance.
 - ii) A simply supported beam 2.5 m long is carrying two point loads of 2 kN and 4 kN at a distance of 1 m & 1.5 m from left end respectively. Draw the SF & BM diagram. 2+6
6.
 - i) What are the assumptions in Simple Bending Theory?
 - ii) A simply supported beam of span 5 m has a cross section 150 mm x 250 mm. If the permissible stress is 10 N/mm^2 , find the maximum intensity of uniformly distributed load it can carry. 2+6
7.
 - i) Write the formula for maximum slope of a cantilever beam with a point load at free end.
 - ii) A cantilever beam of 2 m length having width 100 mm and depth 120 mm is subjected to a point load of 30 kN at its free end. Find the maximum slope and deflection at free end if the Young's Modulus of the beam material is 200 GPa. 1+7

8. i) What is slenderness ratio?
ii) Find the Euler's crippling load for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thick. Take length of the column as 2.3 m and hinged at its both ends. Take $E = 205 \text{ GPa}$. 2+6
9. i) Write down the equation of Pure torsion and explain each term.
ii) A solid circular shaft transmits 100 kW power at 200 r.p.m. Calculate the shaft diameter if twist in the shaft is not to exceed 1° in 2 metres of the shaft length and the shearing stress is limited to 70 N/mm^2 . Take modulus of rigidity $= 8 \times 10^4 \text{ N/mm}^2$. 2+6
10. i) What is Spring index?
ii) A close coil helical spring to carry a load of 250 N, if the mean coil diameter is to be 8 times that of the wire, calculate these diameters. Take maximum shear stress as 70 MPa . 2+6
-

**STRENGTH OF MATERIALS**

Time Allowed: 3 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten):
2×10=20

- i) The strength of a beam depends on –
(a) bending moment (b) its weight ✓
(c) section modulus (d) shape of the section
- ii) If a coiled spring is cut into two halves, stiffness of the spring will be –
(a) double, (b) half, (c) same, (d) quadruple.
- iii) A beam is a structural element subjected to _____ load.
(a) axial, (b) transverse, (c) seismic, (d) none of these.
- iv) If a rod expands freely due to heating, it will develop –
(a) thermal stress, (b) bending stress, (c) compressive stress, (d) no stress.



- v) If diameter of a shaft is doubled the power transmitted capacity will be –
(a) same, (b) either twice or half, (c) four times, (d) eight times. ✓
- vi) Polar moment of inertia of a circular area of diameter 'D' –
(a) $\frac{\pi}{64} D^4$, (b) $\frac{\pi}{32} D^4$, (c) $\frac{\pi}{8} D^4$, (d) —
- vii) A strut is a member which carries –
(a) tensile load, (b) shear load, (c) compressive load, (d) both tensile and compressive load. ✓
- viii) The maximum deflection of a simply supported beam of length 'l' with a uniformly distributed load of 'w' per unit length is –
(a) —, (b) —, (c) —, (d) —
- ix) The shape of shear force diagram for a cantilever with a point load at the free end will be –
(a) rectangular, (b) curve, (c) square, (d) triangular.



- x) The intensity of shear stress at a section is _____ to the distance of the section from the axis of the shaft.
(a) inversely proportional, (b) directly proportional, (c) equal, (d) parallel
- xi) The relationship between Young's modulus (E), Bulk modulus (K) and Poisson's ratio (ν) is given by –
(a) $E = 3K(1 - 2\nu)$ (b) $E = 2K(1 - 2\nu)$
(c) $E = 3K(1 + 2\nu)$ (d) $E = 2K(1 - 3\nu)$
- xii) If diameter of a long column is reduced by 30%, the percentage reduction of Euler buckling load is – (a) 59, (b) 49, (c) 8, (d) 76.
- xiii) The slenderness ratio is the ratio of effective length to least _____.
(a) ultimate load (b) actual length
(c) radius of gyration (d) factor of safety
- xiv) When the helical extension spring is subjected to axial tensile force, the type of stress induced in the spring wire is –
(a) compressive stress, (b) torsional shear stress, (c) tensile stress, (d) bending stress.
- xv) In a thin cylindrical shell subjected to an internal pressure (p), the ratio of hoop stress to longitudinal stress is – (a) $1/2$, (b) 1, (c) 2, (d) 4



GROUP-B

Answer any Five (05) questions.

2. a) Illustrate stress-strain curve for ductile (M.S) material & discuss salient points.
b) The diameter of a water main is 2 m and the thickness of its wall is 20 mm. Determine the circumferential and longitudinal stresses set up in the pipe material when water is introduced in it at a pressure of 1.5 N/mm^2 . Take, Young's modulus (E) = 200 kN/mm^2 .
4+4
3. a) Deduce the formula for the elongation of a uniform cross-sectional bar under the action of its own weight.
b) Find the force required to punch a hole of 20 mm in diameter in a metal plate of 10 mm thickness. If the ultimate strength of the plate metal in shear is 300 N/mm^2 , determine the compressive stress set up in the punch.
5+3



3223770

4.

Find the moment of inertia of the I-section shown in Fig.1 about the horizontal axis passing through the centroid of the section.

8

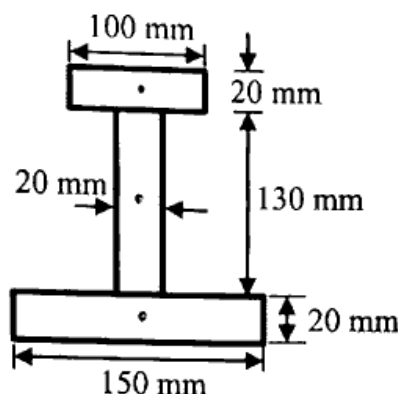


Fig.1

5.

Draw shear force and bending moment diagrams for the beam with loads as shown in the Fig.2. Show all important points in the diagrams.

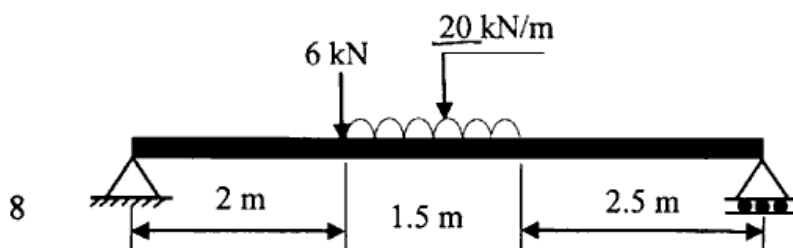


Fig. 2



6. a) What do you mean by pure bending?
 b) Deduce the bending equation: $\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$ with usual notations. 2+6
7. a) A cantilever 2 m long carries a uniformly distributed load of 30 kN per metre run over its entire length. Taking, Young's modulus (E) = 2×10^4 kN/cm² and moment of inertia of the section of the cantilever about the neutral axis = 10,000 cm⁴. Determine, (i) deflection at the free end, (ii) slope at the free end.
 b) What is flexural rigidity? 6+2
8. a) What is the difference between a column and a strut?
 b) A hollow column of 200 mm external diameter and 160 mm internal diameter is used as a column of 4.5 m length. Calculate the Rankine crippling load when the column is pinned at both ends.
 Take allowable stress as 300 MPa and Rankine's constant as 1/7500. 3+5



3223770

9. a) The diameter of a solid shaft is 250 mm. Find out the maximum safe torque which the shaft can transmit if the allowable shear stress is 4600 N/cm^2 and the maximum angle of twist is 0.25° per metre length. Take, Modulus of rigidity (G) = $8 \times 10^6 \text{ N/cm}^2$. If the shaft rotates at 200 r.p.m, what maximum power can be transmitted by the shaft? 6+2
- b) What do you mean by polar modulus?
10. a) Deduce a formula for the equivalent stiffness of helical springs arranged in series connection and parallel connection.
- b) A closed coiled helical spring having the mean coil diameter is 60mm and the diameter of the wire with which the spring is made is 5 mm. The axial load carried by the spring is 200 N. The spring is made of 20 effective turns. Taking, Modulus of rigidity (G) = 80 GPa, determine – 4+4
- (i) maximum shear stress developed in the spring, (ii) deflection of the spring.



<https://www.wbscteonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

JANUARY 2024

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

i. Which point on the stress strain curve occurs after the ultimate point?

- a) Last point b) Breaking point
 c) Elastic limit d) Material limit

ii. The hinged end is also known as _____

- a) Fixed end b) Pinned end
 c) Rigid end d) Free end

iii. The slenderness ratio is the ratio of effective length to least _____

- a) Ultimate load b) Actual length
 c) Radius of gyration d) Factor of safety

iv. _____ is half the circumferential stress.

- a) Hoop stress b) Longitudinal stress
 c) Fluid stress d) Transverse stress

v. What are the units of Polar modulus?

- a) mm^3 b) mm^2
 c) mm d) mm^4

vi. U.D.L stands for?

- a) Uniformly diluted length b) Uniformly developed loads
 c) Uniaxial distributed load d) Uniformly distributed loads

vii. Units of U.D.L?

- a) KN/m b) KN-m
 c) $\text{KN-m} \times \text{m}$ d) KN

viii. At the Point of contraflexure, what is the value of bending moment?

- a) one b) zero
 c) three d) infinity

ix. When SF is zero, the bending moment is _____

- a) Zero b) Maximum
 c) Very difficult to say d) Minimum

x. What is the moment of inertia of a rectangular section about an horizontal axis through C.G?

- a) $bd^3/6$ b) $bd^2/12$
c) $b^2d^2/12$ d) $bd^3/12$

xi. What is the section modulus (Z) for a rectangular section?

- a) $bd^2/6$ b) $a^3/6$
c) BD^3-bd^3 d) D^4-d^4

xii. On bending of a beam, which is the layer which is neither elongated nor shortened?

- a) Axis of load b) Neutral axis
c) Center of gravity d) None of the mentioned

xiii. In cantilever beam the deflection occurs at _____

- a) Free end b) Point of loading
c) Through out d) Fixed end

xiv. The angle of twist can be written as _____

- a) TL/J b) CJ/TL
c) TL/CJ d) T/J

xv. Shear force is unbalanced _____ to the left or right of the section.

- a) Horizontal force b) Vertical force c) Inclined force d) Conditional force

Group-B

2. Draw the stress-strain diagram for a mild steel specimen. Show the salient points in the diagram (8)

3. A cantilever beam AB, 2m long carries a uniformly distributed load of 1.5 kN/m over a length of 1.6 m from the free end. Draw shear force and bending moment diagrams for the beam. (8)

4. Find the moment of inertia of a T-section with flange as 150 mm x 50 mm and web as 150 mm x 50 mm about X-X and Y-Y axes through the centre of gravity of the section. (8)

5. Calculate the maximum torque that a shaft of 225 mm diameter can transmit, if the maximum angle of twist is 1° in a length of 1.5 m. Take $C = 70 \times 10^3 \text{ N/mm}^2$ (8)

6. A steel rod 5 m long and of 40 mm diameter is used as a column, with both end hinged. Determine the crippling load by Euler's formula. Take $E = 200 \times 10^3 \text{ N/mm}^2$ (8)

7. A closely-coiled helical spring of round steel wire 5 mm in diameter having 12 complete coils of 50 mm mean diameter is subjected to an axial load of 100 N. Find the deflection of the spring and the maximum shearing stress in the material. Modulus of rigidity (C) = $80 \times 10^3 \text{ N/mm}^2$ (8)

8. A steam boiler of 800mm diameter is made up of 10 mm thick plates. If the boiler is subjected to an internal pressure of 2.5 MPa. Find the circumferential and longitudinal stresses induced in the boiler plates. (8)

9. Write short notes on the followings (Any four) a) Elasticity b) Malleability c) Fatigue d) Creep e) Poisson's Ratio f) Brittleness (4×2 = 8)

10. Prove that $T/J = C.\theta/L$ for a solid circular shaft. What are the assumptions made in derivation of torsion equation. (5 + 3 = 8)

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for the right answer and 1 mark is allotted for the correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20
 - (i) The stress at which elongation of a material is quite large as compared to the increase in load is known as-
(a) Ultimate point (b) Yield point (c) Elastic limit (d) Rupture point.
 - (ii) For a Poisson's ratio 0.4 for a material, the ratio of the shear modulus of elasticity to modulus of elasticity is- (a) 14/5 (b) 5/7 (c) 7/5 (d) 5/14.
 - (iii) Which of the following increases bending stress in a beam?
(a) Increase in moment of inertia (b) Increase in section modulus (c) Decrease in distance from neutral axis (d) Decrease in moment of inertia.
 - (iv) If a simple supported beam is loaded with a concentrated load W at the centre of the span then its B.M diagram takes the shape of -
(a) Rectangle (b) Parabola (c) Triangle (d) Square.
 - (v) When a beam is subjected to a negative bending moment-
(a) Top fiber is in tension and bottom fiber is in compression (b) Top fiber is in compression and bottom fiber is in tension. (c) Minimum stress in top and bottom fiber (d) Both (a) and (b).
 - (vi) The M.O.I. of a circle of diameter D with respect to the axis which is perpendicular to its diameter and passing through the centre is -
(a) $\frac{\pi}{64} D^4$ (b) $\frac{\pi}{32} D^4$ (c) $\frac{\pi}{32} D^3$ (d) $\frac{\pi}{16} D^3$
 - (vii) In a simple supported beam carrying UDL ' w ' per unit length, the point of contraflexure -
(a) Lies in the centre of the beam (b) Lies at the ends of the beam (c) Depends upon the length of the beam (d) Does not exist.
 - (viii) If a shaft diameter is doubled, its torsional strength increases by-
(a) 2 times (b) 4 times (c) 8 times (d) 16 times.
 - (ix) The buckling load of a column for a given material depends upon-
(a) Poisson's ratio and slenderness ratio (b) Slenderness ratio and modulus of elasticity (c) Poisson's ratio and modulus of elasticity (d) Slenderness ratio and cross sectional area.
 - (x) The ratio of torsional strengths of solid to hollow shafts, both having outside diameter D and hollow having inside diameter $D/2$, is-
(a) 15/16 (b) 1/8 (c) 1/16 (d) 16/15.

- (xi) In the context of beam bending, flexural rigidity is -
 (a) A measure of the beam's resistance to shear stress (b) A measure of the beam's resistance to bending
 (c) A measure of the beam's ability to support compressive loads (d) A measure of the beam's ability to support tensile loads.
- (xii) A column is fixed in one end and other end is free and has a radius of gyration $K = 2.5\text{ cm}$. The actual length of the column is 1.2 m . The slenderness ratio of the column is -
 (a) 96 (b) 80 (c) 90 (d) 120.
- (xiii) If the flexural rigidity of a beam becomes double then the deflection of the beam-
 (a) Increases 2 times (b) Becomes half (c) decreases 4 times (d) remains same.
- (xiv) Four springs each having stiffness k are arranged in series, then the equivalent stiffness will be-
 (a) $4k$ (b) $k/2$ (c) $2k$ (d) $k/4$.
- (xv) Which of the following end condition gives the lowest Euler's buckling strength for a column?
 (a) Both ends fixed (b) Both ends pinned (c) One end fixed, other free (d) One end fixed, other hinged.

GROUP-B

2. (a) Draw the stress-strain curve for M.S material and discuss salient points in the diagram. (3)
 (b) A circular steel bar having three segments is subjected to various forces at different cross sections as shown in fig. 1. The diameter of the first, middle and last section of the bar is 30 mm , 25 mm and 30 mm respectively. Determine the total elongation of the bar. Take $E = 200\text{ GPa}$. [3+5]

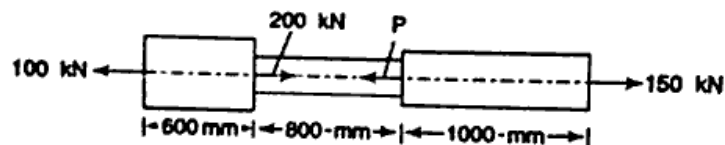


Fig. 1.

3. (a) A bar of 12 mm diameter and 2 m length is subjected to an axial load of 20 kN . The change in diameter is measured as 0.003 mm . If the modulus of rigidity of the bar is 80 GPa , determine (i) Poisson's ratio (ii) modulus of elasticity and bulk modulus.
 (b) No stress is developed in the rails when they are laid at 24°C . If the rails are 32 m long, calculate the stress developed in the rails at 80°C when it is allowed to expand 8 mm . Take co-efficient of linear expansion, $\alpha = 11 \times 10^{-6}/^\circ\text{C}$ and $E = 205\text{ GPa}$. [5+3]
4. Determine the moment of inertia of the composite shape given in the fig. 2 about x-axis and y-axis. [8]

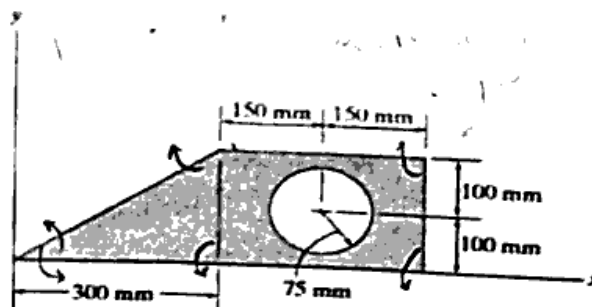


Fig. 2.

5. Draw shear force and bending moment diagrams for the beam as shown in the fig.3. Determine the maximum bending moment and its location. [8]

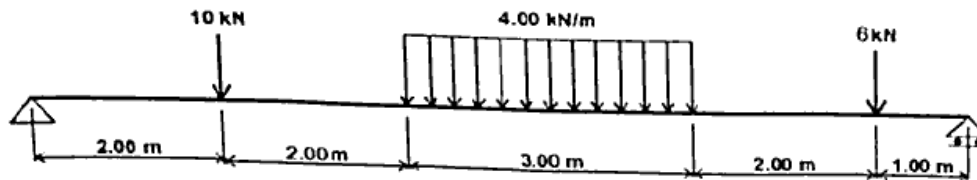


Fig.3.

- 6.(a) What do you mean by section modulus of a beam? (2)
 (b) A cast iron beam of T section as shown in fig.4 is simply supported on a span of 8m. The beam carries a uniformly distributed load of 1.5kN/m length on the entire span. Determine the maximum tensile and maximum compressive stress. [2+6]

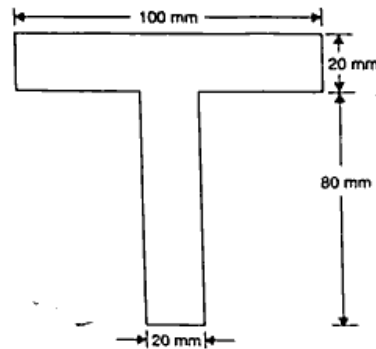


Fig.4.

7. (a) State the four assumptions in theory of bending.
 (b) A beam of uniform rectangular section 20cm wide and 30cm deep is simply supported at its ends. It carries a uniformly distributed load of 9 kN/m run over the entire span of 5 m. If the value of E for the beam material is $1 \times 10^4 \text{ N/mm}^2$, find the slope at the supports and maximum deflection. [2+6]
8. (a) What is torsional stiffness?
 (b) A hollow shaft with inner to outer diameter ratio 3/5 is required to transmit 450kW at 120rpm with a uniform twisting moment. The shear stress in the shaft must not exceed 60 MPa and twist in a length of 2.5m must not exceed 1° . Calculate the least external and internal diameter of the shaft satisfying these conditions. Take $G = 80 \text{ GPa}$. [2+6]
9. (a) Draw the effective lengths of a column for various end conditions. (2)
 (b) A hollow cylindrical column of length 6m having external diameter 200mm and thickness 25mm is fixed at both ends. Determine the ratio of Euler's critical load to Rankine's critical load. Take $E = 80 \text{ GPa}$; allowable stress as 550 MN/m^2 and Rankine's constant as $1/1600$. [3+5]
10. (a) What are the different types of springs? State the functions of the springs.
 (b) A closely coiled helical spring is to carry a load of 500 N. The spring index is 10. If the maximum shear stress in the material of the spring is to be 80 MPa, determine mean coil diameter and wire diameter of the spring. If the stiffness of the spring is 20 N per mm deflection, calculate the energy stored in the spring and the number of turns in the spring. Take $G = 80 \text{ GPa}$ [2+6]